# Variance Retention as a Diagnostic Complement to Persistence-Relative Skill in Multi-Horizon $PM_{10}$ Forecasting

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**Abstract**

Persistence-relative RMSE skill is the standard yardstick for multi-horizon point-forecast evaluation, but a skilled forecast under this criterion can still be operationally uninformative if its predictions collapse toward a near-constant value. We evaluate multi-horizon $PM_{10}$ point forecasts under a rolling-origin protocol with persistence as an explicit baseline, comparing LightGBM and SARIMA across lead times $h \in \{1, 6, 24, 48\}$ hours at an hourly-resolution station in Madrid, Spain. Both models achieve high persistence-relative RMSE skill across all evaluated horizons (LightGBM: 0.92–0.96; SARIMA: 0.89, flat across horizons), an apparent equivalence that does not survive a variance-based reading: LightGBM retains 78–99% of observed variance depending on horizon, while SARIMA retains less than 2% at $h = 1$ and near zero at longer leads. SARIMA's forecasts are thus close to a constant offset, and its RMSE skill reflects error reduction achieved by suppressing amplitude rather than by tracking the target series. We report horizon-wise variance retention alongside RMSE-based skill, and introduce $\text{Skill}_{VP}$, a bounded diagnostic auxiliary that discounts nominal skill by variance retention, to summarize this decoupling in a single horizon-wise quantity. Event-oriented evaluation under fold-recalibrated exceedance thresholds confirms the operational consequence: SARIMA detects approximately 40% of high-$PM_{10}$ events, against 89–91% for LightGBM. Together, these results show that RMSE-relative skill alone can substantially overstate the operational value of a multi-horizon forecast, and that reporting variance retention alongside it is a low-cost, broadly applicable safeguard against this failure mode.

**Keywords:** $PM_{10}$ forecasting; multi-horizon forecasting; rolling-origin evaluation; persistence baseline; forecast skill; variance retention; dynamic fidelity

## 1 Introduction

Forecasts of particulate matter ($PM_{10}$) support operational activities such as public health advisories, short-term compliance planning, and the allocation of monitoring and mitigation resources [Donnelly et al., 2015]. In such settings, model evaluation is often framed around useful lead times: the range of horizons for which forecasts provide actionable information beyond simple references. Multi-horizon forecasting and horizon-wise verification are therefore standard in air quality applications, and persistence remains a common baseline because it is transparent and frequently competitive at short lead times [Murphy, 1992].

A practical complication is that squared-error criteria (MSE/RMSE) dominate both model optimization and the reporting of multi-horizon performance. Under limited predictability, point forecasts that shrink toward a central tendency and dampen fluctuations can reduce squared error by avoiding large misses, even if this also suppresses variability that may be operationally relevant [Gneiting, 2011]. This trade-off is well known in point forecasting under squared loss: improvements in average error can coincide with reduced amplitude, and this tension may become more salient as lead time increases and the conditional distribution of outcomes broadens.

The gap addressed in this paper is empirical and protocol-specific. In multi-horizon $PM_{10}$ forecasting evaluated with rolling-origin protocols, positive persistence-relative RMSE skill can coexist with degraded dynamic fidelity, reflected by a pronounced reduction in predicted variance relative to observed variance at the same horizon. In such cases, a horizon-wise skill curve may suggest that forecasts remain useful at medium leads while the corresponding prediction sequences become under-variable and less informative about episode evolution. Rolling-origin evaluation is central to this diagnostic because it aggregates performance over a sequence of forecast origins spanning changing conditions, thereby exposing horizon-dependent behaviors that may be less visible under a single blocked split [Tashman, 2000].

This study provides a bounded empirical characterization of this decoupling for LightGBM and SARIMA evaluated against a persistence baseline. We report horizon-wise RMSE and persistence-relative skill together with observed and predicted variances and variance retention, and we introduce $\text{Skill}_{VP}$ as a simple diagnostic auxiliary metric to summarize how nominal skill and variance retention jointly behave at each horizon. The pattern in which persistence-relative RMSE skill remains positive while predictive variance is substantially attenuated is sometimes described informally in the forecasting community as "ghost skill"; we use the term here only as shorthand for that specific decoupling and do not adopt it as formal terminology.

**Contributions.** The contribution is intentionally bounded and empirical:

- We provide horizon-wise evidence, under rolling-origin evaluation with persistence as an explicit baseline, that persistence-relative RMSE skill can decouple from dynamic fidelity as measured by variance retention. LightGBM and SARIMA serve as contrasting cases: both are RMSE-skilled relative to persistence, yet they differ sharply in variance retention and in exceedance-detection ability.
- We introduce $\text{Skill}_{VP}$ as a simple diagnostic auxiliary metric that adjusts nominal persistence-relative RMSE skill by variance retention for horizon-wise interpretation, and we report event-oriented complements (recall, precision, flag rate) under fold-recalibrated thresholds as a further check on operational usefulness.

# 2 Related Work and Positioning

This manuscript is positioned as a restrained empirical-diagnostic study in multi-horizon environmental forecasting. It does not aim to advance general forecasting theory or introduce new universal verification metrics. Instead, it operationalizes several established verification ideas in a deterministic $PM_{10}$ case study using a rolling-origin protocol, persistence as an explicit baseline, horizon-by-horizon reading of skill, and an explicit diagnosis of variance retention.

## 2.1 Forecast skill and baseline-aware evaluation

The horizon-wise perspective motivates treating the horizon as the primary unit of interpretation in multi-step evaluation, rather than collapsing performance into a single aggregate number [Ben Taieb et al., 2012]. In applied environmental forecasting, this framing is paired with explicit reference forecasts that provide an operationally meaningful benchmark for interpreting whether a model adds information beyond simple persistence-like behavior [Murphy, 1992].

The present paper adopts this baseline-aware, horizon-wise reading in a deliberately narrow setting: deterministic point forecasts for $PM_{10}$ evaluated under a rolling-origin protocol with persistence as an explicit baseline. Rolling-origin aggregation preserves temporal causality and reduces the optimistic bias that can arise when temporal dependence is insufficiently respected, while also exposing horizon-dependent behavior across changing conditions [Tashman, 2000].

## 2.2 Variance-fidelity diagnostics

A well-known concern in deterministic point forecasting is that squared-error optimization can encourage shrinkage toward conditional means, which may suppress predictive variability as lead time increases [Gneiting, 2011]. In environmental and hydrological verification, this issue connects to antecedents that explicitly separate different components of performance, including diagnostics that track variability ratios (e.g., the variability component of KGE-style decompositions) and adjacent discussions of oversmoothing and amplitude damping [Gupta et al., 2009, Kling et al., 2012].

The ingredients used here are classical: a baseline-relative skill quantity and a variance ratio. Letting Skill($h$) denote persistence-relative RMSE skill at horizon $h$ and letting $\sigma^2_{\hat{y}}(h)$ and $\sigma^2_y(h)$ denote the horizon-wise predicted and observed variances over the rolling-origin verification set, we define

$$\mathrm{Skill}_{VP}(h) = \mathrm{Skill}(h)\,\frac{\sigma^2_{\hat{y}}(h)}{\sigma^2_y(h)}, \tag{1}$$

used here in a bounded form (see Section 3.7). Skill$_{VP}$ is positioned only as an auxiliary diagnostic complement that down-weights nominal persistence-relative RMSE skill when variance retention is low; its precise scope and boundaries are stated in Section 3.7.

### 2.3 Contribution statement

The strongest contribution of this paper is not conceptual novelty, but the empirical combination of a $PM_{10}$ case study with rolling-origin evaluation, persistence as an explicit operational baseline, horizon-by-horizon performance reading, and an explicit variance-retention diagnosis. In the reported results, LightGBM and SARIMA serve as contrasting cases: both are RMSE-skilled relative to persistence, but SARIMA provides the clearer extreme in which persistence-relative RMSE skill remains positive while predictive variance is near-collapsed. This contrast motivates interpreting skill curves jointly with variance retention when discussing horizon-wise usefulness, without claiming that such decoupling must occur universally.

## 3 Methodology

### 3.1 Forecasting setting and multi-horizon notation

Let $y_t$ denote the observed $PM_{10}$ concentration at time index $t$. At each forecast origin $t$, a point forecasting model issues multi-horizon predictions $\hat{y}_{t+h|t}$ for horizons $h \in \{1, \ldots, H\}$. Evaluation is performed separately for each horizon to avoid conflating short- and long-lead behavior, since both error characteristics and the operational interpretation of forecasts can change substantially with lead time.

### 3.2 Rolling-origin evaluation

The empirical analysis was conducted on an hourly $PM_{10}$ series from the Casa de Campo monitoring station in Madrid, Spain, covering the full calendar year 2023 (2023-01-01 00:00 to 2023-12-31 23:00; 8,760 observations). The data were obtained from the European Environment Agency (EEA) Air Quality Download Service. Table 1 summarises the study site, data coverage, and evaluation protocol.

We adopt a rolling-origin (walk-forward) evaluation protocol to approximate sequential operational use. For each origin $t$ in an evaluation window, the model is trained using only information available up to $t$ and predictions are generated for all horizons. For each horizon $h$, verification aggregates over the set of valid origins $\mathcal{T}_h$ with targets $y_{t+h}$ available, yielding horizon-wise sample sizes $N_h = |\mathcal{T}_h|$. This protocol avoids look-ahead and exposes the evaluation to changes in conditions across origins [Tashman, 2000].

### 3.3 Reference forecast: persistence

Persistence is used as an explicit baseline forecast. For each origin $t$ and horizon $h$, the persistence prediction is

$$\hat{y}^{(p)}_{t+h|t} = y_t. \tag{2}$$

All skill statements in this paper are made relative to persistence and are reported horizon-wise.

### 3.4 Empirical models: LightGBM and SARIMA

We report results for two empirical point-forecasting models. The first is a gradient-boosted decision tree model (LightGBM), trained independently for each horizon $h$ using lag features, rolling-window statistics,

Table 1: Summary of the study site, data coverage, and evaluation protocol.

| Item | Value |
|---|---|
| Study site / station | Casa de Campo (Madrid, Spain) |
| Period | 2023-01-01 00:00 to 2023-12-31 23:00 |
| Temporal resolution | 1 hour |
| Number of observations | 8,760 |
| Data source | EEA Air Quality Download Service |
| Forecast horizons ($h$) | 1, 6, 24, 48 h |
| Rolling-origin folds | 5 (expanding window, initial 50%) |
| Holdout split | 80/20 chronological |
| Models compared | LightGBM; SARIMA |
| Baseline | Persistence |

and temporal covariates [Ke et al., 2017]. LightGBM is included as a strong, widely used nonlinear predictor in applied forecasting.

The second is SARIMA, a classical time-series model fitted on the raw normalized $PM_{10}$ series at each rolling-origin fold using a fixed order $(p,d,q)(P,D,Q)_s = (1,0,1)(1,0,1)_7$ and recursive multi-step forecasting [Box et al., 2015]. SARIMA is included not as a comprehensive statistical benchmark, but as a comparator that can exhibit qualitatively different dynamic-fidelity behavior under the same rolling-origin evaluation and persistence-relative skill framing.

## 3.5 Horizon-wise RMSE and persistence-relative RMSE skill

For model $m$ at horizon $h$, the rolling-origin RMSE is

$$\mathrm{RMSE}_m(h) = \sqrt{\frac{1}{N_h} \sum_{t \in \mathcal{T}_h} \left( \hat{y}^{(m)}_{t+h|t} - y_{t+h} \right)^2}. \tag{3}$$

Persistence-relative RMSE skill is defined as

$$\mathrm{Skill}_{\mathrm{RMSE}}(h) = 1 - \frac{\mathrm{RMSE}_m(h)}{\mathrm{RMSE}_p(h)}. \tag{4}$$

Positive values indicate lower RMSE than persistence at horizon $h$.

## 3.6 Dynamic fidelity via variance retention

To characterize amplitude preservation, we compute horizon-wise sample variances over the same evaluation set $\mathcal{T}_h$:

$$\mathrm{Var}_{\mathrm{obs}}(h) = \mathrm{Var}(\{y_{t+h} : t \in \mathcal{T}_h\}), \qquad \mathrm{Var}_{\mathrm{pred}}(h) = \mathrm{Var}\Big(\{\hat{y}^{(m)}_{t+h|t} : t \in \mathcal{T}_h\}\Big). \tag{5}$$

Variance retention is reported as a percentage,

$$\mathrm{VR}(h) = 100 \times \frac{\mathrm{Var}_{\mathrm{pred}}(h)}{\mathrm{Var}_{\mathrm{obs}}(h)}. \tag{6}$$

Values substantially below 100% indicate that predictions are under-variable relative to observations at the same horizon.

## 3.7 Skill$_{VP}$ as a diagnostic auxiliary metric

RMSE-based skill and variance retention capture different aspects of forecast behavior: the former summarizes average squared deviations, while the latter summarizes amplitude preservation. To characterize

Table 2: Canonical horizon-wise results under rolling-origin evaluation (E1). VR denotes variance retention (%), and $\text{Skill}_{VP}$ is a diagnostic auxiliary adjustment of persistence-relative RMSE skill by variance retention.

| | LightGBM | | | | SARIMA | | | |
|---|---|---|---|---|---|---|---|---|
| $h$ | RMSE | Skill | VR (%) | $\text{Skill}_{VP}$ | RMSE | Skill | VR (%) | $\text{Skill}_{VP}$ |
| 1 | 0.335 | 0.963 | 78.2 | 0.753 | 0.984 | 0.890 | 1.8 | 0.016 |
| 6 | 0.726 | 0.919 | 80.9 | 0.743 | 0.985 | 0.890 | 1.2 | 0.011 |
| 24 | 0.375 | 0.958 | 97.6 | 0.935 | 0.989 | 0.889 | 0.3 | 0.003 |
| 48 | 0.381 | 0.957 | 98.7 | 0.945 | 0.991 | 0.889 | 0.1 | 0.001 |

horizons where persistence-relative RMSE skill coexists with low variance retention, we define $\text{Skill}_{VP}$ as a variance-preserving diagnostic auxiliary:

$$\text{Skill}_{VP}(h) = \text{Skill}_{\text{RMSE}}(h) \times \min\left(1, \frac{\text{Var}_{\text{pred}}(h)}{\text{Var}_{\text{obs}}(h)}\right). \tag{7}$$

The $\min(\cdot)$ bound prevents inflation when predictions are over-variable. This construction is intentionally simple and is scoped narrowly: $\text{Skill}_{VP}$ is a horizon-wise diagnostic complement to RMSE-based skill, not a candidate replacement for it, and it makes no claim about probabilistic calibration or decision-theoretic optimality. Its sole purpose is to flag, at a glance, horizons where nominal skill and amplitude preservation diverge.

# 4 Results

## 4.1 E1: Canonical horizon-wise results under rolling-origin evaluation

The central result is a sharp divergence in dynamic fidelity between two models that both achieve high persistence-relative RMSE skill. Table 2 and Figure 1 show that, while LightGBM and SARIMA are similarly RMSE-skilled relative to persistence across all evaluated horizons, their variance-retention profiles are qualitatively different: LightGBM preserves a substantial fraction of the observed variability, whereas SARIMA produces near-constant forecasts that retain almost none.

Specifically, both models achieve high persistence-relative RMSE skill across $h \in \{1, 6, 24, 48\}$ (LightGBM: 0.963, 0.919, 0.958, 0.957; SARIMA: 0.890, 0.890, 0.889, 0.889). Interpreted in isolation, these values could be read as evidence that both models remain useful relative to persistence across all evaluated lead times. This reading does not survive the variance-retention diagnostic. LightGBM retains a substantial fraction of observed variance (VR = 78%, 81%, 98%, 99% at $h = 1, 6, 24, 48$), consistent with a model that preserves meaningful amplitude variation in its predictions. SARIMA, by contrast, exhibits near-collapsed predictive variance despite remaining RMSE-skilled (VR = 1.8%, 1.2%, 0.3%, 0.1%), indicating that its predictions are close to a constant offset and that persistence-relative RMSE skill in this case masks severe loss of dynamic fidelity.

$\text{Skill}_{VP}$ makes this decoupling explicit per horizon by attenuating persistence-relative RMSE skill when variance retention is low (Figure 2, left). For LightGBM, $\text{Skill}_{VP}$ remains relatively close to RMSE-based skill at longer horizons where variance retention is near-complete ($\text{Skill}_{VP} = 0.935$, 0.945 at $h = 24, 48$), with more pronounced attenuation at shorter horizons (0.753, 0.743 at $h = 1, 6$). For SARIMA, the same high RMSE-relative skill levels translate into near-zero $\text{Skill}_{VP}$ values at every horizon (0.016, 0.011, 0.003, 0.001), making the dynamic-fidelity loss explicit in a single summary quantity.

Importantly, the LightGBM results do not support a strong narrative of long-horizon variance collapse within the currently evaluated range. Variance retention increases with horizon for LightGBM, reaching 98–99% at $h = 24, 48$. The stronger diagnostic signal is model-dependent: SARIMA provides the clearer extreme case, while LightGBM illustrates that high RMSE skill can coexist with comparatively plausible amplitude preservation.

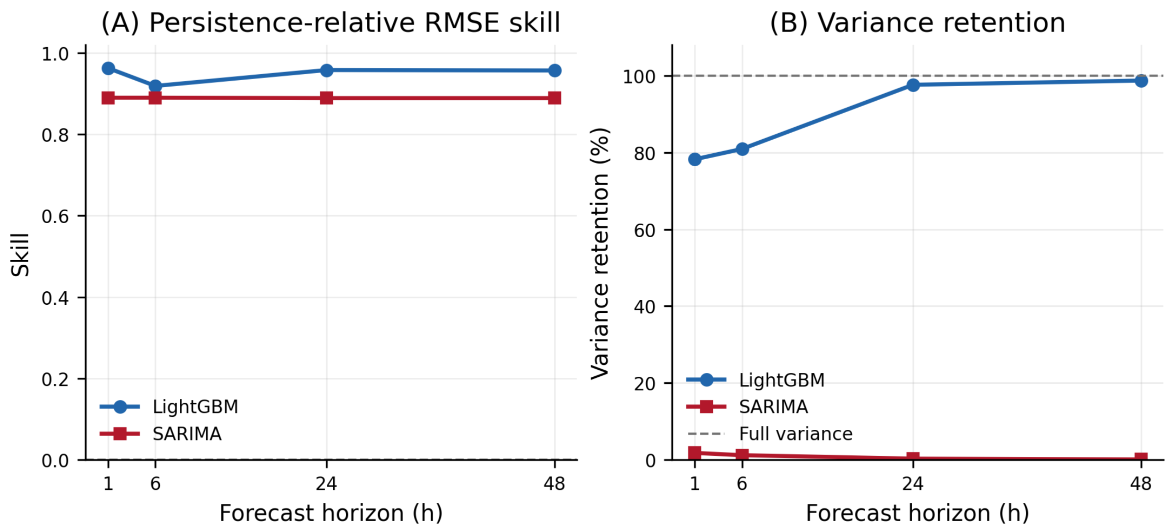


Figure 1: Rolling-origin horizon-wise results for LightGBM and SARIMA. *Left*: persistence-relative RMSE skill. *Right*: variance retention (%). Both models appear similarly skilled under RMSE; variance retention reveals a sharp divergence in dynamic fidelity.

## 4.2 E2: Protocol robustness (rolling-origin vs. blocked holdout)

To assess whether the variance-retention contrast is specific to rolling-origin aggregation, we report an analogous blocked-holdout evaluation. The holdout results support the same qualitative pattern. LightGBM retains a substantial fraction of observed variance under holdout (VR = 55.6%, 56.7%, 94.6%, 92.9% with $\text{Skill}_{VP} = 0.532, 0.523, 0.910, 0.894$ for $h = 1, 6, 24, 48$). SARIMA remains RMSE-skilled but exhibits very low variance retention under holdout as well (VR = 3.8%, 2.6%, 0.6%, 0.1% with $\text{Skill}_{VP} = 0.033, 0.022, 0.006, 0.001$). These results confirm that the model-dependent decoupling between persistence-relative RMSE skill and dynamic fidelity is not an artifact of the rolling-origin aggregation.

## 4.3 E3: Event-oriented complement

As a supportive operational complement, we evaluate exceedance detection at the P75 threshold under fold-recalibrated thresholds (one threshold per rolling-origin fold, fitted on that fold's training data). The results reinforce the variance-retention contrast: LightGBM achieves recall of 0.89–0.91 and precision of 0.80–0.93 across horizons, while SARIMA achieves recall of approximately 0.40 and precision that degrades from 0.79 at $h = 1$ to 0.30 at $h = 48$. SARIMA's flag rate ($\approx$0.40) shows that its predictions rarely exceed the fold-recalibrated threshold, consistent with the near-constant forecasts implied by near-zero variance retention.

These event-oriented results are treated as supportive diagnostics. Recall alone can be trivially high under threshold saturation—when forecasts remain above a permissive threshold at most time steps—and should therefore be interpreted jointly with precision, flag rate, and variance retention. Under fold-recalibrated thresholds, both recall and precision reflect meaningful predictive discrimination for LightGBM, while SARIMA's event detection is limited by its inability to predict concentration peaks.

Overall, E1–E3 jointly support a bounded, model-dependent conclusion: persistence-relative RMSE skill can remain positive while dynamic fidelity differs markedly across models, and variance retention together with $\text{Skill}_{VP}$ provides a practical diagnostic view of this divergence for horizon-wise interpretation.

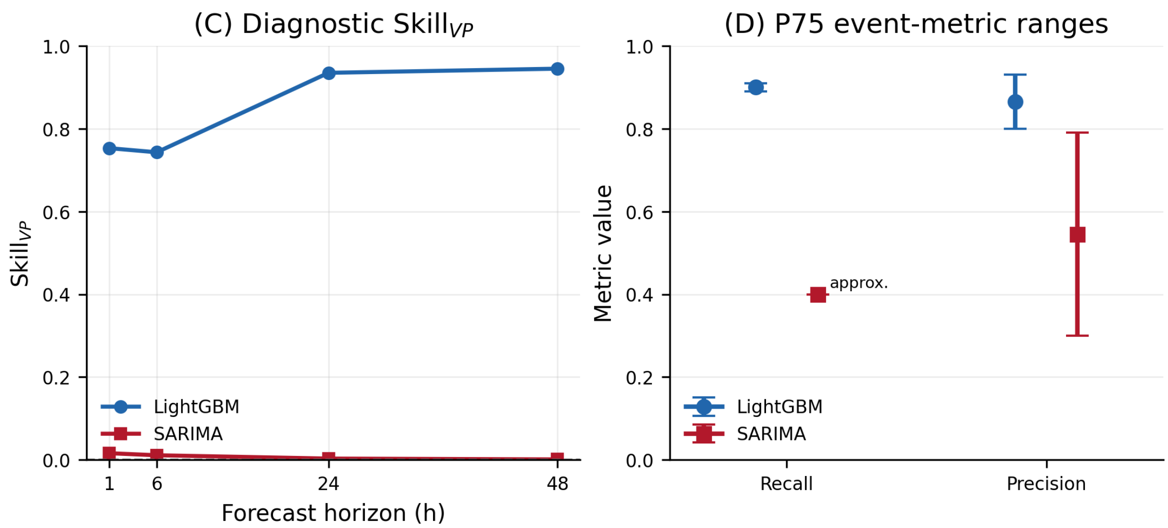


Figure 2: Diagnostic complements for horizon-wise interpretation. *Left*: Skill$_{VP}$, which adjusts persistence-relative RMSE skill by variance retention. *Right*: reported across-horizon ranges for exceedance recall and precision at the P75 threshold under fold-recalibrated evaluation; SARIMA recall was reported approximately.

## 5 Discussion

The evidence in this manuscript supports a diagnostic, horizon-wise interpretation of multi-horizon $PM_{10}$ point forecasts under rolling-origin evaluation with persistence as an explicit baseline. Across the evaluated horizons, both LightGBM and SARIMA remain positively skilled relative to persistence under RMSE-based summaries. However, the variance diagnostics reveal that similar persistence-relative RMSE skill can coincide with sharply different levels of dynamic fidelity: LightGBM is a moderate case with substantial variance retention, whereas SARIMA is an extreme case in which predictive variance is near-collapsed. This contrast is consistent with the view that error-based skill alone can mask severe losses of amplitude in the predicted sequence, and that horizon-wise interpretation changes materially when variance retention is examined alongside RMSE-relative skill.

Skill$_{VP}$ is useful as a diagnostic auxiliary metric precisely because it makes this divergence explicit and comparable across horizons without requiring additional probabilistic infrastructure. The LightGBM results do not support a strong long-horizon variance-collapse narrative; variance retention at longer horizons is comparatively high, and Skill$_{VP}$ adjustments are moderate. The SARIMA results, by contrast, provide clear evidence that RMSE-relative skill can coexist with near-zero variance retention, and the event-oriented evaluation confirms that this has direct operational consequences: a model that appears useful under RMSE skill may fail to detect concentration peaks when its predictions are near-constant.

Event-oriented evaluation must be handled cautiously. Recall alone may be trivially high under threshold saturation and does not by itself demonstrate predictive discrimination or operational usefulness. In this study, fold-recalibrated thresholds prevent artificial inflation of event base rates, and recall is reported jointly with precision and flag rate. Under these conditions, the event metrics support the same conclusion as the variance diagnostics: LightGBM retains operational usefulness across horizons, while SARIMA's operational value is limited despite its RMSE-relative skill.

## 6 Limitations

This study is intentionally limited in scope. The empirical evidence is anchored in a single station, a single calendar year, and five rolling-origin folds; external validity to other pollutants, temporal resolutions, monitoring sites, or longer evaluation windows has not been established. The results are restricted to the evaluated horizons $h \in \{1, 6, 24, 48\}$ and to two model classes (LightGBM, SARIMA) with specific, fixed configurations; neither model was tuned to compete for state-of-the-art accuracy, since the goal is to characterize a decoupling pattern rather than to benchmark model performance. $\text{Skill}_{VP}$'s scope is stated in Section 3.7 and is not repeated here; the metric has not been validated outside the present case study, and its behavior under regimes with strong seasonality or structural breaks remains untested.

# 7 Conclusion

In this $PM_{10}$ case study evaluated under rolling-origin protocols with persistence as an explicit baseline, both LightGBM and SARIMA achieve high persistence-relative RMSE skill across all evaluated horizons, yet their variance-retention profiles diverge sharply. LightGBM retains a substantial fraction of observed variance and achieves meaningful exceedance detection, while SARIMA produces near-constant forecasts that retain almost no variability and detect only 40% of high-$PM_{10}$ events. Reporting variance retention alongside RMSE skill, and using $\text{Skill}_{VP}$ as a diagnostic auxiliary metric, offers a practical way to make this divergence explicit for horizon-wise interpretation under rolling-origin evaluation, without implying a universal replacement for established verification practices.

# Data and Code Availability

The analysis code, configuration files, and evaluation scripts used in this study are available at `https://github.com/fedeg-umh-es/varret-pm10-paper`. Raw $PM_{10}$ data were obtained from the European Environment Agency (EEA) Air Quality Download Service (`https://aqportal.discomap.eea.europa.eu/`) and are publicly available under the EEA standard data policy.

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